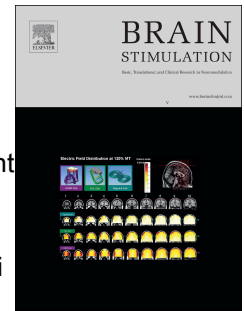


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Title:

Effects of Prefrontal Theta-Burst Stimulation on Brain Function in Treatment-Resistant Depression: A Randomized Sham-Controlled Neuroimaging Study

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Abstract

Background: Theta-burst transcranial magnetic stimulation (TBS) can quickly modulate brain activity and can be used to treat treatment-resistant depression (TRD). Whole-brain analytical research has revealed that left high-frequency PFC rTMS modulates brain activity in anterior cingulate cortex (ACC) and the fronto-cingulo-temporal circuit. We aimed to investigate whether the prefrontal TBS's antidepressant mechanisms involve these regions.

Methods: We designed a randomized, double-blind, sham-controlled neuroimaging study to investigate different mechanisms of TBS [i.e., continuous TBS (cTBS) and intermittent TBS (iTBS)]. This study included 56 TRD patients [67.8%: women; mean age (SD): 45.5 (10.6) years], who were randomly assigned to one of four groups [A: right prefrontal cTBS; B: left prefrontal iTBS; C: combined right prefrontal cTBS and left prefrontal iTBS; and D: sham-TBS]. We measured standard uptake values (SUV) of cerebral glucose metabolism, at rest, both before and after ten daily treatment sessions.

Results: Group B and Group C had more responders than Group A and Group D ($\chi^2 = 9.161$, $p = 0.027$). In Group A, the SUV was significantly increased in the ACC and medial PFC (mPFC) after treatment, while the SUV was decreased in the right temporal cortex ($p < 0.05$, FWE-corrected for multiple comparisons). In contrast, in Group B, SUV was decreased in the ACC and mPFC and increased in the bilateral temporal cortex. Group C also had increased SUV in the bilateral temporal cortices.

Conclusion: Instead of observing inhibitory or facilitatory effects on the targeted prefrontal cortex, we found that the two-week prefrontal TBS protocols primarily modulated the fronto-cingulo-temporal circuit.

Clinical trial registration

Registered name: Therapeutic Effects of Theta-Burst Stimulation on the Severity of Depression and Cognitive Functions in Major Depression

URL: https://upload.umin.ac.jp/cgi-open-bin/ctr_e/ctr_view.cgi?recptno=R000013859

Registration number: UMIN000011856

Keywords: theta-burst stimulation; glucose metabolism; theta oscillation; anterior cingulate cortex; prefrontal cortex; major depression

Introduction:

Repetitive transcranial magnetic stimulation (rTMS) of the prefrontal cortex (PFC) is current treatment option for adults with treatment-resistant depression (TRD) that has been approved by the U.S. FDA. Considerable evidence has supported the antidepressant efficacy of daily prefrontal rTMS specifically aimed at the left PFC, which has been widely used in clinical settings [1]. However, only 30-40% of depressed patients achieve symptomatic remission after being treated with rTMS [2]. In one recent study, an altered form of the rTMS protocol, namely theta-burst stimulation (TBS), was proposed; this method features such advantages as short administration duration, use of low intensity stimulus pulses, and powerful effects on synaptic plasticity [3]. Previous studies have shown that TBS is a safe intervention, even when using parameters that are higher than the suggested stimulation threshold [4]. Furthermore, an increasing amount of research has found this method to be an effective treatment option for patients with TRD [5-8]. Nevertheless, its antidepressant mechanisms with regard to PFC are still unclear.

With regard to TBS mechanisms, most published studies have focused on the primary motor cortex (M1) in both healthy control subjects and patients with various movement disorders [9]. Intermittent TBS (iTBS) produces long-term potentiation-like effects over M1, while continuous TBS (cTBS) provides a long-term depression-like reduction of cortical

excitability on it [10]. As more research on TBS is completed, significant variations have been reported in TBS responses over M1 across individuals [9]. The effect of TBS on other cortical regions has remained largely unexplored and may vary from those observed in M1. For example, *Huang and Mouraux* reported that not all the after-effects of cTBS were inhibitory [11]. Furthermore, *Hamada et al.* analyzed 56 healthy subjects and reported that their responses to inhibitory and excitatory TBS protocols varied significantly [12]. That particular study revealed that only 25% of the subjects demonstrated inhibitory responses to cTBS and facilitatory responses to iTBS (31%: cTBS facilitatory and iTBS inhibitory, 27%: both facilitatory, and 17%: both inhibitory). Therefore, the theoretical mechanisms (i.e., cTBS as inhibitory and iTBS as excitatory) may not completely explain the prefrontal TBS effects on TRD.

Despite of a lack of prefrontal TBS studies on brain networks of depression, review data has pointed out that high-frequency PFC rTMS tended to increase regional brain activity, while low-frequency tended to decrease the activity [13]. In addition, PFC rTMS was shown to modulate not only regional but also remote brain regions (e.g., ACC, temporal cortex, limbic circuit, insula) [13]. Paul et al. reported that left high-frequency PFC rTMS modulates brain activity in the fronto-cingulate circuit [14]. Likewise, a recent resting-state functional network study demonstrated that left PFC high-frequency rTMS mainly modulated anterior

cingulate cortex (ACC) connectivity and a specific network involving PFC, ACC, temporal cortex, striatum, and insula [15]. Through studying D2 receptors by positron emission tomography, Cho et al. demonstrated that high-frequency PFC rTMS induced a significant reduction of D2-receptor binding potentials mainly in midline prefrontal regions, including ACC and PFC [16]. Therefore, it is plausible that prefrontal TBS may also involve the above-mentioned brain regions.

Furthermore, a key element of non-invasive brain stimulation is rhythm (“theta”, in the case of TBS). Researchers have found that brain stimulation frequency tuning might target underlying oscillatory brain activity, subsequently enhancing stimulation specificity by selectively targeting brain neurons/circuits with certain oscillatory frequencies [17] (for further details, please refer to [17]). With regard to theta rhythm, the electroencephalographic theta signal mainly correlates with frontal midline distribution (i.e., ACC and PFC) in humans, as well as proactive and reactive cognitive control processes [18, 19]. Findings from a pioneer study showed that, in researching couplings of the electroencephalographic theta signal and whole-brain glucose metabolism, positive correlations were mainly observed in the fronto-temporal regions [20], while the strongest couplings of resting theta rhythm were found to come mainly from the frontal midline (e.g., ACC and PFC) [20].

Currently, neuroimaging studies exploring prefrontal TBS mechanisms for treating TRD are still lacking. Therefore, this sham-controlled study aimed to exploratorily investigate the brain effects of different prefrontal TBS protocols using ^{18}F -fluorodeoxyglucose positron emission tomography (^{18}F -FDG-PET). Notably, the study was part of a large randomized double-blind sham-controlled trial of TBS and the antidepressant efficacy and cognitive effects of TBS had been reported previously [8, 21]. As discussed above, we hypothesized that prefrontal TBS may modulate brain activities of the fronto-cingulo-temporal regions.

As a secondary objective, we explored the core antidepressant mechanism of prefrontal TBS by examining brain changes in patients with antidepressant responses to a two-week TBS treatment regimen.

Materials and Methods

Subjects

For this study, we recruited adult patients with a DSM-IV diagnosis of recurrent major depressive disorder, which was diagnosed through a thorough medical history check and the Mini International Neuropsychiatric Interview (MINI) (American Psychiatric Association, 1994). Furthermore, the recruited subjects had already failed to respond to at least two antidepressant treatments given with typically adequate dosages and durations in the current depressive episode. The subjects' current depressive episode was required to have a Clinical Global Impression Scale (CGI-S) score of at least 4 and a total Hamilton Depression Rating Scale-17 (HDRS-17) score of at least 18 [22], which allowed us to recruit patients with moderate to severe depression. We excluded patients if they had a history of psychotic disorders; bipolar I or II disorders; substance abuse or dependence; personality disorders (based on DSM-IV criteria); any medical history of major systemic illness or neurological disorders (e.g., seizure, stroke, post-brain surgery, traumatic brain injury); brain implants (neurostimulators) or cardiac pacemakers, or if they were pregnant. This study was performed pursuant to the Declaration of Helsinki and was approved by the relevant Ethics Review Committee, and we obtained written informed consent from all participants. The study was part of a large randomized double-blind sham-controlled trial that aimed to investigate the antidepressant and cognitive effects of TBS in Taiwan and was preregistered in the University Hospital Medical Information Network Clinical Trials Registry (Registration number: UMIN000011856), one of the largest clinical trial registries in Asia and a member of the World Health Organization (WHO) registry network.

Study procedures

We took a detailed psychiatric and medical history of the participating subjects, who also underwent a diagnostic interview and brain structural imaging by MRI at baseline for screening and PET image co-registration. We randomized a total of 56 eligible patients over two years (age range: 23-64 years) 1:1:1:1 into four TBS groups (Groups A, B, C, and D) and provided two weeks of a different TBS treatment for each group (patients had to stay on their original medication regimen throughout the study). To ensure equal sample sizes among the four groups over time, we used a computerized random number generator with the block randomization method (block size of eight), which was carried out by an independent research assistant. However, in order to maintain adequate blinding, neither the investigators nor the patients knew the block size. The number of participants in each group was determined prior to initiating the study using the Power and Sample Size Program (version 3.0.43) [23] pursuant to the findings of previous studies [8, 24, 25]. During the treatment, one TBS session was scheduled daily for 5 consecutive days, with a total of 10 sessions over the two-week treatment period. The entire study procedure was double-blinded, with all patients and staff blind to the treatment assignment.

With regard to the heterogeneous nature of major depression, we measured baseline refractoriness levels using the Maudsley Staging Method (MSM) [26]. The MSM is a point-based staging model that incorporates three main factors to cover various dimensions of treatment resistance: treatment (i.e., number of antidepressant treatment failures and whether

augmentation or electroconvulsive therapy has been used), symptom severity, and duration of the current episode.

Depressive symptoms at baseline (Week 0, prior to TBS treatment) and at the end of the two-week TBS treatment (one hour after the last session of the TBS stimulation, Week 2) were assessed with HDRS-17 [22] by blinded study personnel who had been forbidden from accessing any data of the treatment sessions. Patients were considered as having responded if they had at least a 50% reduction in their baseline HDRS-17 by the end of Week 2. In general, the present study focused on the before-versus-after findings of brain glucose metabolism in response to the two-week TBS treatment, as well as in the subjects whose depressive symptoms responded well to TBS.

Theta-burst stimulation parameters and session procedures

We performed the TBS sessions with a Magstim Rapid2 stimulator (Magstim Company, Ltd), using brain-navigation computer software and an infrared system (Brainsight, Rogue Research, Inc., Montreal, QC) to guide the coil to target the dorsolateral prefrontal cortex (DLPFC) over the patient's head (for each subject's brain MRI scan) [27]. Although some studies found that modified TBS protocols (e.g., modified cTBS: three pulses at 30-Hz) may produce more consistent inhibitory after-effects than standard parameters [28] and could modify brain regions beyond motor cortex [29], we adopted standard TBS parameters

(3-pulse 50-Hz bursts, every 200 ms at 5 Hz, and an intensity of 80% active motor threshold) [30]. Patients in Group A received continuous TBS to the right DLPFC (cTBS: 1800 pulses/session); Group B patients received intermittent TBS to the left DLPFC (iTBS: a 2-s train of bursts repeated every 10 s, 1800 pulses/session); Group C patients received a combination of iTBS and cTBS (iTBS-1800 + cTBS-1800 pulses/session; in randomly assigned order starting from intermittent TBS or continuous TBS); and patients in Group D received sham TBS (bursts given as cTBS or iTBS, randomly assigned; 1800 pulses/session) with the coil set at 90 degrees to the skull [31]. Notably, TBS has been shown to have dose-dependent effects. For example, by investigating motor evoked potentials (MEP), Nettekoven et al. found that iTBS blocks of 1800 pulses over primary motor cortex increased MEP amplitudes more than that of 600 pulses in human subjects [32]. Further details can be found in the supplemental material.

Imaging procedures

We obtained MR images with a 3.0 GE Discovery 750 whole-body high-speed imaging device, which provided high-resolution structural T1-weighted images. Two volumes of ¹⁸F-FDG PET scans (i.e., pre- and post-TBS treatment; the brain acquisition time for each PET volume was 15 minutes) of at rest glucose utilizations were acquired using a PET/CT scanner (Discovery VCT; GE Healthcare, USA) in 3D brain mode in the morning (0900 h to

1200 h). For each PET volume, a 15-minute PET scan in a resting state was acquired 45 minutes after an intravenous (IV) injection of approximately 222 MBq of ^{18}F -FDG, after which the subjects had to stay awake in a dimly lit room. More information about MRI and PET acquisition can be found in the supplemental material.

Preprocessing and analyzing the PET data

We analyzed the PET data using Statistical Parametric Mapping version 8 software (SPM8) implemented in MATLAB 7.1 (The Mathworks Inc., Sherborn, MA, USA). We further created a group-specific MRI-aided ^{18}F -FDG template to normalize each subject's PET images [33, 34]. The smoothed (full width at half maximum [FWHM] = 8 mm) and normalized PET images in the standardized brain space were then utilized for further analysis. We adopted the parametric whole-brain standardized uptake value (SUV), which is a validated semi-quantitative method, by dividing the concentration of radiotracer activity in the normalized PET image (MBq/kg) by the FDG dose at the time of injection (MBq) divided by the body weight (kg) [35].

Whole-brain (voxel-wise) analysis of PET data

Our primary outcome measures were the before-versus-after SUV changes for each TBS group, which we analyzed using paired *t*-tests. The significance thresholds were set at a

voxel-level uncorrected value of $p < 0.005$ and cluster-level family-wise error (FWE)-corrected value of $p < 0.05$. To ascertain the nature of TBS's antidepressant mechanisms, before-versus-after SUV changes for responders in Group B and Group C were also studied with paired t -tests; this step was not done for Group A or Group D due to too few responders. Furthermore, we compared the between-group SUV changes using two-way ANOVA, with time (i.e., the 1st and 2nd SUV images) as the within-subject factor, TBS groups as the between-subject factor, and HDRS-17 changes as a covariate. We then analyzed the group-by-time interaction and the main effects of group and time that were found to be significant at $p < 0.05$ after performing voxel-level family-wise error (FWE) correction.

Our secondary outcome measures were voxel-wise partial correlations, which were performed to study any potential correlation between improved depressive symptoms (% HDRS change in 2 weeks) and the baseline whole-brain SUV map after covarying for age, gender, and total gray matter in order to determine a value that could predict TBS antidepressant effects with baseline neuronal activity. The results were considered significant at a cluster-level corrected value $p < 0.005$.

Region of interest (ROI) analysis of PET data

To visualize the individual results observed in the whole-brain analysis of each patient, we performed ROI analysis [36]. The mean SUV values in the ACC, medial PFC, left

temporal cortex, and right temporal cortex were extracted from unsmoothed SUV images in the standard stereotactic space by using PMOD version 3.0 (PMOD Technologies, Zurich, Switzerland) as described in previous studies [37, 38]. We then reported the normalized glucose metabolism for each ROI (values were normalized by dividing the mean SUV of each ROI by the global mean SUV). We delineated the ROIs using an automated anatomical labeling template (AAL template [[39]]) in order to prevent bias from inter- or intra-rater reliability issues arising from manual delineation.

Statistical methods

All statistical analysis of demographic and clinical data was performed using SPSS 16.0 (SPSS Inc.). We further adopted one-way ANOVA (or Student's t-test) and Fisher's chi-square test (or Yate's correction) to compare the continuous and categorical variables among groups, respectively. Paired t-tests were used to compare the continuous variables before and after treatment. P-values less than 0.05 were considered statistically significant, and the least significant difference (LSD) was used for *post-hoc* analyses.

Results

Demographic data and treatment outcomes

Fourteen subjects for each group were recruited for investigating the brain effects of different prefrontal TBS protocols. Notably, most of the patients (52/56, 92.9%) were from the same study population as reported previously [8]. For the present neuroimaging analysis, we recruited four more subjects (one for each group). The four subjects were also non-responders in response to the two-week brain stimulation treatment, so the final results of the antidepressant efficacy for different TBS protocols did not differ from the previous report [8]. To avoid duplication of the clinical effects, here we only highlighted some very important demographic and clinical variables. Baseline age, gender, and HDRS-17 scores did not differ significantly among the four groups. The mean (SD) of treatment refractoriness determined by MSM among the four groups did not differ [A: B: C: D = 10.3(1.9) : 9.1(1.6) : 9.0(1.7) : 8.9 (1.6); F -value=2.081, p =0.114]. Other clinical variables, such as psychiatric comorbidities and medications, were also similar among the four groups (Table S1). Regarding the changes of HDRS-17 scores between Week 0 and Week 2, the ANOVA analysis revealed that the four groups showed significant differences (Table 1, ANOVA for four groups, F -value=5.160; p =0.003), with the active groups (Group A : -24.3% ; Group B : -42.6% ; Group C : -52.5%) showed considerable improvements in HDRS-17 scores, while the sham group (Group D: -18.2%) showed little improvement (Post-hoc: Group B<Group D, Group C<Group D, Group

C<Group A). Furthermore, the ANOVA analysis for the active groups showed significant differences among the three groups ($F=4.467$; $p=0.018$; post-hoc: Group C<Group A). Group B and Group C also had considerably more responders than Group A and Group D (Table 1, four-group analysis, $\chi^2=9.161$, $p=0.027$). Among the active groups, the chi-square analysis showed no significant differences among the three groups ($\chi^2=5.534$, $p=0.063$). All of the recruited subjects both completed the neuroimaging part and had two scans (i.e., before and after the brain-stimulation treatment).

Results of regional brain glucose metabolism

No differences were observed in the baseline PET activities among the four groups. As for the primary outcome measures of the before-versus-after PET findings following ten sessions of TBS, right cTBS (Group-A) significantly increased SUV (standard uptake values) in two clusters: the first cluster included the ACC and medial PFC (peak coordinates: $x, y, z = -6, 30, 36$; 2985 voxels, $P_{\text{corrected}}=0.003$, FWE-corrected for multiple comparisons), while the second included the posterior cingulate cortex and precuneus (peak= $-10, -50, 16$; 2977 voxels, $P_{\text{corrected}}=0.020$) but demonstrated decreased SUV in the right temporal cortex (peak= $50, -12, 2$; 767 voxels, $P_{\text{corrected}}=0.019$)(Fig. 1A). In contrast, left iTBS (Group B) considerably decreased SUV in a cluster that included the ACC and medial PFC (peak= $0, 40, 20$; 1226 voxels, $P_{\text{corrected}}=0.004$) and contralateral PFC (peak= $38, 28, -4$; 735 voxels, $P_{\text{corrected}}=0.019$).

Furthermore, left iTBS (Group B) increased SUV in the left temporal cortex (peak= -62, -28, 6; 1460 voxels, $P_{\text{corrected}} = 0.006$) and right temporal cortex (peak= 64, -16, 2; 2634 voxels, $P_{\text{corrected}} = 0.007$) (Fig. 1B). A combination of right cTBS and left iTBS (Group C) increased SUV in many brain regions, including the right temporal cortex (peak= 58, -22, 14; 2033 voxels, $P_{\text{corrected}} < 0.001$), left temporal cortex (peak= -54, -28, 2; 1604 voxels, $P_{\text{corrected}} < 0.001$), parietal cortex (peak= 26, -8, 72; 260 voxels, $P_{\text{corrected}} = 0.028$), and precuneus (peak= 4, -6, 54; 273 voxels, $P_{\text{corrected}} = 0.016$) but decreased SUV in the cerebellum (peak= -2, -54, -32; 516 voxels, $P_{\text{corrected}} = 0.018$) (Fig. 1C). The sham group (Group D) had decreased SUV in the superior parietal cortex (peak= 20, -52, 62; 214 voxels, $P_{\text{corrected}} = 0.033$) (Fig. 1D). Two-way ANOVA analysis did not reveal a significant relationship between group and time. Meanwhile, subsequent ROI analyses corresponded to the whole-brain analytical results, indicating that the brain regions with the most significant changes after TBS treatment were mPFC, ACC, and bilateral temporal cortex (Fig. S1).

Relationship between regional brain glucose metabolism and depression improvement

Our correlation analysis indicated that higher pretreatment hippocampal activities were related to improved depressive symptoms following a two-week TBS treatment regimen (FWE-corrected $p < 0.005$), regardless of whether the correlation was performed for all subjects or just those who received active TBS (Fig. 2, Fig. 2A for all, and Fig. 2B for active

treatment only). No such findings were found in the sham group. Furthermore, we exploratorily examined the before-versus-after PET results in the responders and found that TBS responders in Group B had decreased SUV in the right insula (peak coordinate=44, -20, 4; 446 voxels, $P_{\text{corrected}}=0.003$) and an increase in the cerebellum (peak coordinate=56, -56, -42; 220 voxels, $P_{\text{corrected}}=0.025$). Likewise, TBS responders in Group C demonstrated decreased SUV in the right insula (peak coordinate=40, -32, 14; 385 voxels, $P_{\text{corrected}}=0.008$) but an increase in the left temporal cortex (peak coordinate=-56, -64, 12; 348 voxels, $P_{\text{corrected}}=0.001$) and right temporal cortex (peak coordinate=46, -16, 12; 290 voxels, $P_{\text{corrected}}=0.003$)(Fig. 3). Meanwhile, we did not observe any such findings in the insula in non-responders, but non-responders did demonstrate significant brain activity changes in the cerebellum, parietal cortex, thalamus, and left temporal cortex ($P_{\text{corrected}} < 0.05$) (Fig. S2).

Discussion

To the best of our knowledge, this double-blind sham-controlled neuroimaging study is the first to describe the central mechanisms of different prefrontal TBS protocols. The principal pattern of TBS-related brain changes over the PFC for 2 weeks generally did not reveal observable inhibitory or facilitatory effects on the targeted DLPFC but primarily changed ACC, medial PFC, and temporal cortex activities (Fig. 1). This study provides supporting evidence that right prefrontal cTBS for 2 weeks significantly increased SUV in the ACC and medial PFC and that the left prefrontal iTBS decreased SUV in the ACC and medial PFC (Fig. 1). Such findings were consistent with a recent study, in which Chung et al. found opposite changes in TMS-evoked theta oscillations following prefrontal iTBS (increase) and cTBS (decrease) in healthy subjects [40].

Furthermore, protocols involving iTBS (i.e., Group B and Group C) were found to be more effective, and both demonstrated increased brain activities in the bilateral temporal cortices, while cTBS alone (i.e., Group A) had no significant antidepressant effect and demonstrated decreased activities in the right temporal cortex (Fig. 1). Responders in Group B and Group C consistently showed a unique pattern of metabolic changes in the insula at Week 2. Altogether, these findings indicate that prefrontal TBS mainly modulates the fronto-cingulo-temporal circuit, and the persistence of functional changes in the ACC and PFC at Week 2 may not be a prerequisite for the TBS's antidepressant effects. The absence of

metabolic changes in the ACC and medial PFC by a combination of iTBS and cTBS may be a cancelling out of the metabolic activities. In addition, the antidepressant effects and the metabolic changes of bilateral temporal cortices associated with Group C here might be due to dosing effects (Group C: 3600 pulses; Group B: 1800 pulses) and need further studies to investigate, since a dose-dependent antidepressant effect of TBS has been suggested [41].

Prior studies have shown that TBS focused at the PFC was capable of modulating brain activity, not only in the stimulated cortical area but also in remote and contralateral brain areas [42, 43]. Our findings agreed with those of such previous studies, as well as further revealed that prefrontal TBS treatment primarily modulated the fronto-cingulo-temporal circuit in depression. Such finding was not unexpected, given that prefrontal rTMS has been found to modulate the fronto-cingulate circuit [13-15]. Tik et al. recently investigated the effects of prefrontal high-frequency rTMS in a placebo-controlled large-sample approach (60 healthy subjects), revealing that stimulation of left PFC specifically modulated ACC and its connections with temporal cortex, insula, and striatum [15]. The present sham-controlled PET study extended the finding to TBS and consistently showed that TBS could also modulate ACC and the fronto-cingulo-temporal circuit. Baeken et al. investigated the brain effects of accelerated high-frequency rTMS, which is an aggressive form of rTMS providing 20 rTMS sessions over 4 days [44]. They found that responders to the high-frequency rTMS had a significant decrease in a part of ACC (i.e., subgenual ACC). Likewise, Liston et al. studies

functional connectivity in 17 depressed patients, finding that prefrontal rTMS normalized depression-related subgenual hyperconnectivity in the default mode network [45]. Although it seems both iTBS and high-frequency rTMS involved a modulation of ACC activity, the exact locations of ACC are different (i.e., TBS: the dorsal part; rTMS: the ventral part of ACC). Such findings suggest the antidepressant effects of prefrontal TBS and rTMS come from different brain mechanisms. However, future research directly comparing brain mechanisms between TBS and rTMS is warranted.

Of particular note, the negative observation on the targeted DLPFC after two weeks of TBS treatment does not necessarily mean that iTBS or cTBS has no inhibitory or facilitatory effects on the DLPFC. Instead, this finding indicates that direct inhibitory or facilitatory TBS effects on the targeted prefrontal cortex may be involved in the initial stage of TBS treatment, while the long-term effects of prefrontal TBS treatment (i.e., 2 weeks) may eventually surpass the already established inhibitory or facilitatory TBS effects indicated by existing motor cortical studies [10].

Furthermore, we found that the two-week prefrontal iTBS protocol (Group B) and cTBS protocol (Group A) primarily changed the ACC and medial PFC activities. In fact, the ACC and medial PFC are among the brain regions where strong theta signals can be detected at rest [18, 19]. When awake, humans' theta signal is primarily associated with a frontal midline distribution and is connected to proactive and reactive cognitive control processes [18, 19].

The simultaneous application of both EEG and FDG-PET enabled Pizzagalli et al. to discover that theta oscillations at rest significantly correlated with glucose metabolism in the medial PFC, ACC, PCC, precuneus, and temporal cortex [20], most of which we observed in the present study (Fig. 1). Furthermore, the finding of this prefrontal “theta-burst” stimulation (TBS) study on the ACC and medial PFC provided indirect evidence supporting that one of the key effects of non-invasive brain stimulation may be the result of the applied “rhythm” that entrains oscillatory brain activity and manipulates the entrained neurons/circuits. Our finding was in line with a recent TBS-EEG study, in which the investigators found that prefrontal TBS mainly modulated theta oscillations and cTBS and iTBS had opposite effects on the theta measures [40]. However, since we did not apply EEG, further studies are warranted.

The present study found increased activities in both sides of the superior temporal cortex in Group B and Group C, which seems to suggest that the brain effects of iTBS are stronger than those of cTBS. The temporal cortex, particularly the superior temporal gyrus, is involved in emotional perception and social cognition [46, 47] and plays a crucial role in the depression-related pathway, which consists of both the amygdala and prefrontal cortex [36, 47]. Most of the present study’s findings were located in the superior temporal gyrus (Fig. 1A, 1B, and 1C). Previous researchers have found that MDD patients demonstrated correlations with impaired brain functions in the superior temporal gyrus [48, 49]. Tik et al. using

resting-state functional MRI also demonstrated that prefrontal high-frequency rTMS stimulation specifically modulated a functional network including ACC, PFC, superior temporal gyrus, and insula [15]. As shown in the present study, most of the brain regions were found to be engaged in the TBS's central mechanisms. Given that the functional network mainly involves in cognitive control and negative affect [50], it is plausible that prefrontal high-frequency rTMS and iTBS may be used to specifically target these structures involved in cognitive and affective domains such attention, cognitive control, social cognition, and anhedonia. However, it warrants further studies to investigate. For example, a meta-analysis of functional MRI studies reported that the superior temporal gyrus is one of the most consistently identified regions involved in the pathophysiology of MDD [48]. Takahashi et al. further reported that both current MDD and MDD in remission had a significantly reduced volume of bilateral superior temporal cortices [49]. Furthermore, left temporal cortical activation may not be a key factor for the antidepressant success of TBS as increased temporal cortical activity was also observed in iTBS non-responders (Fig. S2A). With regard to MDD, a previous study has found that the right-side superior temporal gyrus volume was more related to the severity of depressive symptoms than the left side [49].

The central mechanism of TBS may differ from that of conventional 10-Hz rTMS. Based on the findings related to the locations of the temporal cortex, we previously reported that following successful rTMS treatment, the significantly elevated metabolism in the left middle,

but not superior, temporal cortex decreased [27]. In another study, an abnormally higher activation was found in the bilateral middle/inferior temporal gyrus when depressed patients were exposed to facial expressions depicting negative emotions [51]. In addition, the brain activity changes in the superior temporal cortex are not the result of the auditory effects of TBS treatment, since the right temporal cortex activity decreased after the 2-week cTBS treatment (Fig. 1A).

Our findings revealed that insula activity only significantly decreased in the TBS responders; furthermore, it was only consistently observed in the iTBS group and the iTBS/cTBS combination group (Fig. 3), suggesting that functional insula decreases play a vital role in the TBS's antidepressant responses. We found the paradigms involving iTBS were significantly more effective than the sham TBS, which were in line with previous rTMS research showing the most effective rTMS parameters seemed to include prefrontal high-frequency rTMS [52]. The insula is a complex brain structure with various cortical connections in the frontal, parietal, temporal, and cingulate cortices [53] and has even been considered as a part of the temporal lobe [54]. The insula is one of the structures in the proposed depression circuit of cortico-subcortico-limbic dysregulation and has been categorized in the attention-cognition domain of depression [55]. Iwabuchi et al. applied prefrontal iTBS to a group of healthy subjects (n=27) and found a single iTBS session was capable of altering the fronto-insular network through alterations of excitatory/inhibitory

interactions, [56]. Furthermore, another study has found that prefrontal iTBS can modulate the neural activities of the prefrontal cortex and insula; however, this is not the case with cTBS alone [4]. According to some studies, the fronto-cingulate-insular circuit may be responsible for the attention-cognition domain of depression [55], while our results suggest that the insula may be the principal antidepressant target of TBS, albeit indirectly through fronto-cingulo-temporal modulation. In contrast, non-responders also demonstrated significant brain activity changes in the cerebellum, parietal cortex, and thalamus. The exact causes for the findings remain unknown; however, our previous PET study had reported that TRD patients have more impaired brain activities in parietal cortex, precuneus, thalamus, and cerebellum than healthy subjects [27]. Whether the activity changes are related to TBS's treatment effects or an exacerbation of TRD-related brain functions warrant further studies.

Hippocampal theta, which originates from the medial septal area in the basal forebrain and is associated with arousal and memory formation, is among the most studied rhythms [57]. Theta rhythm participates in episodic memory by coordinating hippocampal input from other brain regions [58]. Similarly, we found in this study that increased activity in the hippocampus prior to TBS treatment resulted in better antidepressant efficacy from the TBS treatment (Fig. 2). This finding can serve as another indirect piece of evidence to support the theory that "theta rhythm" is a key participant in prefrontal TBS effects.

This study has some limitations that should be mentioned at this point. First, our sample size of 56 subjects was not large and the sample size in the sub-analyses was even smaller. Additional large-scale studies are needed to directly compare the differences among the various TBS protocols and to validate the role of insula. Second, we did not apply electroencephalography to directly measure theta. However, the entrainment of the theta-related brain regions by TBS may not be directly measurable at the cortical level using electroencephalography. For example, by using electroencephalography in a TBS study on healthy subjects, *Noh et al.* found no change of the theta band in response to a TBS session; instead, the change was predominantly observed in the beta band [59]. In the future, studies using neuroimaging tools that can visualize whole-brain rhythm patterns would be helpful. Second, the motor threshold of all patients was measured from the right first dorsal interosseous muscle. Since interhemispheric asymmetry of motor cortical excitability may exist in patients with major depression [60], the antidepressant efficacy of right prefrontal cTBS may have been affected. However, our results were in line with a recent meta-analysis, which reported that unilateral iTBS was significantly more effective than unilateral cTBS in the treatment of major depression [25]. Finally, although the present PET study found that TBS mainly modulate the fronto-cingulo-temporal circuit, we did not apply resting-state functional MRI. Such studies in the future are needed to validate the findings.

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Conclusion

This double-blind, sham-controlled study is the first to investigate the mechanisms of different prefrontal TBS protocols in the treatment of medication-resistant major depressive disorders. In this study, we discovered that the mechanisms of two-week prefrontal TBS protocols mainly involve modulation of the fronto-cingulo-temporal circuit.

Table 1. Demographic data and clinical variables for neuroimaging analysis

	A cTBS	B iTBS	C iTBS+cTBS	D Sham	F/ χ^2	P
Age	48.8 (10.1)	42.8 (11.0)	43.9 (11.3)	46.2 (9.3)	1.281	0.291
Female	10	7	11	10	2.947	0.400
HDRS-17	24.3 (5.7)	23.1 (3.8)	25.7 (5.0)	24.4 (2.5)	0.779	0.511
Responders/ Non-Responders	3 / 11 (21.4%)	5 / 9 (35.7%)	9 / 5 (64.3%)	2 / 12 (14.3%)	9.161	0.027*

Note: This neuroimaging study was part of a large randomized trial and most of the subjects (52/56, 92.9%) were the same as reported previously (for clinical effects in details, please refer to the publication[8]); iTBS: Intermittent Theta-Burst Stimulation; cTBS: Continuous Theta-Burst Stimulation; HDRS-17, 17-item Hamilton Depression Rating Scale; Responders were defined as those with at least a 50% reduction in depression scores after two weeks of TBS treatment.

* p-values less than 0.05 were considered statistically significant. ** p <0.005

Figure Legends

Figure 1. Voxel-wise whole-brain analysis demonstrates the significant changes of glucose metabolism before and after the two-week treatment in the following four groups: (A) Group A: cTBS at the right dorsolateral prefrontal cortex (RDPFC), (B) Group B: iTBS at the left dorsolateral prefrontal cortex (LDPFC), (C) Group C: a combination of cTBS at RDPFC and iTBS at LDPFC, and (D) Group D: sham treatment. Significant thresholds for each pair *t*-test were set at a voxel-level family-wise error (FWE)-corrected level of $p < 0.05$. Contrast bars denote *t*-values.

Figure 2. Pretreatment glucose metabolism was positively associated with improved depressive symptoms following two weeks of TBS treatment for (A) all patients and (B) patients who received active TBS, with crossings showing individual global maximum cluster under $p < 0.05$ corrected for multiple comparisons (A: 294 voxels, $p = 0.014$, global maximum at Talairach (xyz) coordinates of 20, -28, -16; B: 878 voxels, $p = 0.003$, global maximum at Talairach (xyz) coordinates of 22, -26, -16).

Figure 3. Voxel-wise whole-brain analysis demonstrates the significant changes of glucose metabolism in patients who responded to two-week iTBS (A) or iTBS+cTBS (B) treatment, with crossings showing the individual global maximum under a voxel-level family-wise error (FWE)-corrected level of $p < 0.05$. Contrast bars denote *t*-values.

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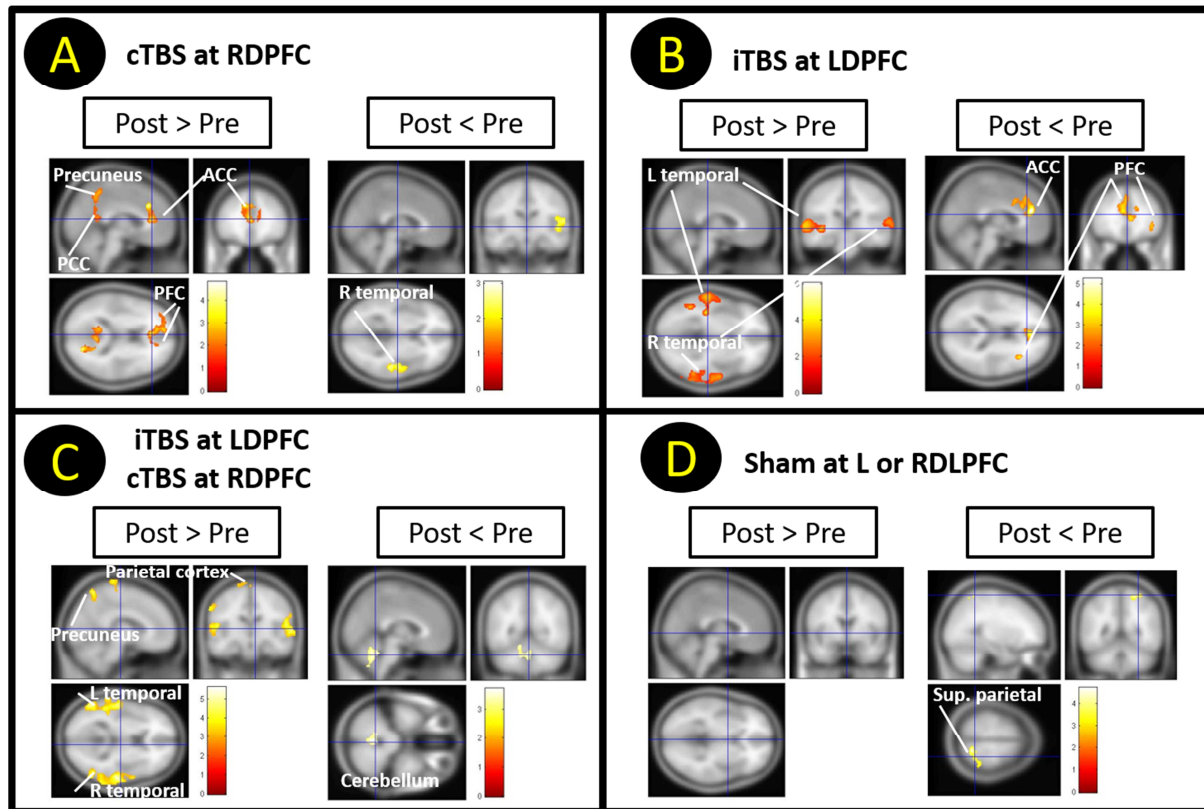
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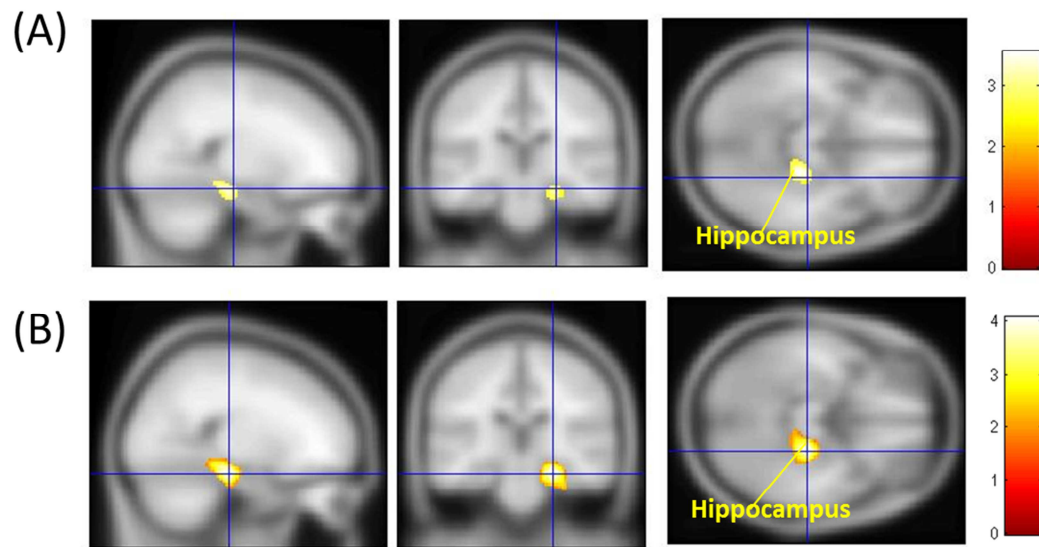
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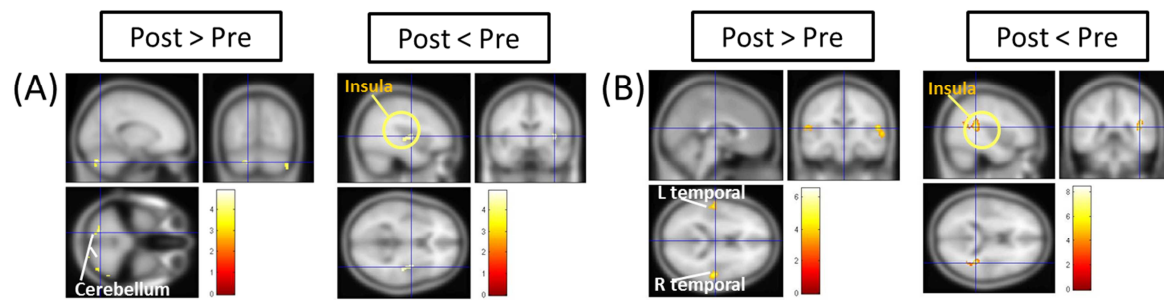
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Highlights:

1. Central mechanisms of prefrontal theta-burst stimulation (TBS) remain elusive.
2. ¹⁸F-FDG-PET scans before and after different forms of TBS were done for refractory depression.
3. The current sham-controlled randomized study found that prefrontal TBS protocols primarily modulated the fronto-cingulo-temporal circuit.